Wan Adzhar Faiq Bin Adzlan

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Experience

AI/ML Engineer | Sandisk | May 2025 – Present

- Architected and solo built a full-stack **Optimization platform using Constraint Programming (OR-Tools)** to compute wafer inventory allocations, saving USD300k+ annually, reducing planning runtime from 2 hours to 5 minutes, and replacing manual Excel-based workflows. Tools: **OR-Tools, FastAPI, React, Typescript, TailwindCSS, Docker, RabbitMQ, PostgreSQL, Jenkins**
- Backend & Agentic RAG lead for an AI chat platform used by 100+ engineers for process Q&A and equipment troubleshooting. **Architected the chat orchestration layer and orchestrator–subagent architecture, enabling the platform to route requests across specialized subagents, MCP servers, and Agentic RAG workflows**; reviewed pull requests and introduced unit tests into CI workflow for TDD. Tools: **LangGraph, MinIO, FastAPI, RabbitMQ, Docker, Kubernetes, vLLM, Single Store, Jenkins**
- Fine-tuned Vision Language Models (**Qwen-2.5-VL**) & cross-encoders for engineering document Q&A/retrieval e.g., flowcharts, images, schematics, etc. on a H200 GPU cluster and built synthetic evaluation datasets using Vision Language Models.
- Set up a Ray Cluster on on-premise VMs with auto-scaling to serve embedding models and enable efficient Ray Data/vLLM/Docling document processing pipelines. Processed and ingested 25k internal engineering documents, achieving 86% HitRate@K. Tools: **Docling, Qwen3 Embeddings, Ray Data/Serve, Single Store, WatchDog, Grafana, Prometheus**

Data Scientist | Maybank | February 2024 – April 2025

- Built Dataiku pipelines using **Facebook Prophet for anomaly detection** to identify large, irregular deposit transactions; enabled targeted retention campaigns that retained MYR18m+ from the treatment group and redirected funds to internal products.
- Built ETL pipelines and **feature engineered 20+ new features related to Auto-Loan/Hire Purchases for a Retail Loans Feature Store.** Increased Precision for loan product take-up by 20% in the department's recommender/Next Best Offer model after extensive Feature Selection tests. Tools: **Hive/HDFS, Spark, Teradata SQL, Dataiku Scenarios, Python, SHAP.**

Senior Data Scientist/Data Scientist | INVOKE | October 2022 – January 2024

- Productionized a **resume parser REST API using *chatgpt-3.5-turbo-1106*** and Langchain to autofill and classify new user registration forms. Outperformed a *chatgpt-4* alternative, reducing latency and cost through prompt engineering. Tools: **OpenAI API, LangChain** (**PydanticOutputParser**), **FastAPI,** AWS (**Lambda, S3, API Gateway**), **GitHub Actions, PyTest.**
- **Built and deployed engagement prediction/product take-up propensity models** for digital marketing ads using Azure Databricks and MLflow; validated impact via A/B testing, achieving a 10% lift in Facebook Ads CTR.
- Built & optimized PySpark ETL pipelines and **built Streamlit dashboards for digital marketing data,** providing a centralised repository of industry benchmarks instead of manual reconciliation on Facebook Ads Manager. ETL used **MongoDB (NoSQL) Aggregation Pipelines, Azure Databricks (Batch processing, Job Scheduler, PySpark**) and **Azure Blob Storage**, while deployment used **Azure App Services, GitHub Actions, PyTest.**
- Lead and developed a *PRN* (State Election's) Simulation/Intelligence Dashboard which uses R(Raking) and rule-based algorithms to predict seat winnability & fence-sitter swings. **226/245 seats predicted accurately (15% increase in accuracy since *GE15*).**

Commercial & Research | INVOKE | September 2021 – September 2022

- Project managed 4 surveys and reported findings to stakeholders from various industries (FMCG, Fintech, Academia).

Personal Projects & Open Source

Open-Source Contributor | Ray, vllm, Malaysia AI | August 2023 – Present

- Provisioned GPU infrastructure for PR validation using Terraform, shell scripts & Runpod.
- Added native Anthropic Messages API support to **Ray Serve LLM**, integrating vLLM request/response protocols, streaming, error handling, ingress routing, and deployment configuration. GitHub
- Extended Ray's LLM batch benchmarking with **vLLM** engine metrics and request-latency analysis (Time per Output Token, Time to First Token, etc.), including percentile statistics and Ray Data overhead estimation. GitHub
- Optimized **llm-compressor** subgraph tracing using *O(1)* dictionary lookups (16.5× speedup on a 100K-target benchmark). GitHub
- Prepared datasets, finetuning and benchmarking scripts (Recall@K/Precision@K) for a Malaysia Reranker model (Mistral architecture) trained on 4x A100 GPUs on Azure Spot VMs. The 64M, 191M & 474M parameter models were added to Malaya, a Malaysian NLP package. Tools: **Weights & Biases, HuggingFace/Transformers, PyTorch, DeepSpeed, FlashAttention2.**

LLM Hackathon Winner | AI Tinkerers Hackathon (sponsored by YTL & SUPA) | October 2024

- **Led a team of 5 for an LLM-as-a-Judge hackathon and won 1st place out of 30 teams.**
- Performed knowledge distillation by finetuning a smaller 7 billion parameter Malaysian Mistral and DeBERTaV2 model for factual/logical inconsistency classification and reasoning ability.
- Generated cost savings by achieving a 73% F1 score on both synthetically generated Malay/English datasets (only 7~% lower than *OpenAI 4o-mini's* F1 score). Tools: **weave, trl (SFTTrainer, QLoRa, PeFT), Huggingface/Transformers, PyTorch.**

Education & Certificates

- Bachelor's Degree, Economics – University of Edinburgh, Scotland (May 2021)